Roll No. | | | | | | | |

ODD END TERM EXAMINATION DECEMBER - 2024

Name of the Course: **M.Tech., ME (CAD/CAM)**
Name of the Paper: **Research Methodology and IPR**

Semester: **IIIrd**
Paper Code: **(MME 301)**

Time: 3 Hours

Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	**(20 marks)**	CO5, CO2, and CO1
(a)	What types of inventions are eligible for patenting, and which matters are considered non-patentable?	
(b)	How can you elaborate the research approaches? Differentiate between the Research Methods versus Methodology.	
(c)	What is the meaning of research? What are the objectives of the research and name its types?	
Q2	**(20 marks)**	CO6, CO2, and CO1
(a)	Discuss Breakthrough Innovation and Incremental Innovation by quoting real time examples.	
(b)	Explain the research process and draw its flow diagram neat and clean.	
(c)	Describe the following: • Descriptive versus Analytical Research • Applied versus Fundamental Research • Qualitative versus Quantitative Research • Conceptual versus Empirical Research	
Q3	**(20 marks)**	CO1, CO3, and CO2
(a)	What is oral presentation? What are merits and demerits of oral presentation?	
(b)	Define sampling in details. Give its all classifications with brief description.	
(c)	Give the light on the followings: • Formulating the research problem • Extensive literature survey • Development of working hypotheses • Preparing the research design	
Q4	**(20 marks)**	CO1, CO5, and CO2
(a)	What is a Geographical Indication? What is a "generic" geographical indication? Why do geographical indications need protection?	
(b)	Elaborate the research problem in detail and what are the techniques involved in defining a research problem?	
(c)	Explain to the good research and what are the main criteria of good research?	
Q5	**(20 marks)**	CO4, CO2, and CO1
(a)	Using a Flow chart, explain the important steps involved in the process of Copyright Registration.	
(b)	How research methodology and IPR will help to engineers and scientists to do a right and ethical research work?	

(c)	What is the need of research methodology and IPR to the academics, society and country? Give a case study where research methodology is important.	